

QUEBEC CITY SAINTE-FOY, QC, Canada

WMO: 713920

Lat: 46.7803N

Lon: 71.2875W

Elev: 300

StdP: 14.54

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-6.6	-21.0	1.8	-9.8	-16.1	2.3	-5.7	23.4	17.8	21.0	19.7	6.3	250	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	84.3	69.0	81.5	67.4	78.8	66.0	72.4	80.1	70.5	77.4	68.8	75.4	7.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.9	111.2	76.1	68.2	104.7	74.6	66.4	98.3	72.7	36.2	80.2	34.6	77.4	33.2	75.6	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	17.4	14.6	DB	-17.0	89.4	5.3	2.8	-20.8	91.4	-23.9	93.0	-26.9	94.6	-30.8	96.6	(4)
(5)			WB	-18.2	75.5	5.4	2.2	-22.0	77.1	-25.2	78.4	-28.2	79.7	-32.0	81.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	12.6	16.3	25.7	39.5	53.9	62.5	68.1	66.5	58.6	45.6	33.6	20.4		
	DBStd	21.00	11.89	10.42	10.07	7.55	7.46	6.47	5.10	5.17	6.85	7.11	8.10	10.45		
	HDD50	4815	1159	944	754	322	41	1	0	0	8	175	493	918		
	HDD65	8658	1624	1364	1218	764	353	123	26	44	214	603	942	1383		
	CDD50	1924	0	0	0	8	162	376	562	512	266	37	1	0		
	CDD65	292	0	0	0	0	9	47	123	91	22	0	0	0		
	CDH74	2015	0	0	0	5	128	366	817	549	149	1	0	0		
	CDH80	412	0	0	0	0	25	87	177	98	25	0	0	0		
	WSAvg	6.2	6.8	6.8	7.0	7.3	7.1	5.8	5.1	4.9	5.0	6.1	6.3	6.6		
	PrecAvg	49.8	3.8	3.3	3.6	3.2	4.0	4.5	4.8	5.0	3.9	4.4	4.6	4.6		
(15) Precipitation	PrecMax	64.9	6.3	7.6	5.9	6.1	7.7	7.2	7.9	8.8	8.6	6.7	7.2	8.7		
	PrecMin	38.9	1.7	0.2	0.8	0.9	1.3	1.6	2.2	1.3	1.4	1.5	2.0	1.9		
	PrecStd	5.8	1.4	1.8	1.1	1.1	1.7	1.4	1.5	2.1	1.6	1.3	1.3	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.7	41.3	52.6	68.8	83.5	86.7	88.5	86.3	83.3	70.9	58.5	43.9		
		MCWB	40.2	38.3	43.6	54.2	63.5	69.7	72.9	71.4	70.1	61.4	53.5	42.5		
	2%	DB	37.0	36.7	45.4	61.6	77.4	82.4	84.5	82.7	78.2	64.8	51.3	38.9		
		MCWB	35.6	33.8	38.7	47.8	60.2	66.8	70.2	68.7	67.3	57.2	47.4	37.2		
	5%	DB	33.5	34.4	41.4	55.6	72.6	78.1	81.5	79.8	74.1	59.7	47.4	36.3		
		MCWB	32.3	32.5	36.6	45.0	58.6	64.5	68.0	67.2	64.9	54.7	44.0	34.6		
	10%	DB	29.5	31.9	38.2	51.1	67.5	74.4	78.6	76.9	70.1	56.0	44.0	33.8		
		MCWB	28.1	30.2	34.4	42.5	56.1	62.7	66.4	66.1	62.8	51.4	41.0	32.4		
	0.4%	WB	40.3	38.7	46.5	57.7	68.2	72.6	75.4	73.8	72.4	63.2	54.1	42.6		
		MCDB	40.7	40.3	50.3	67.0	77.7	82.3	84.5	82.7	79.9	67.9	57.1	43.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	34.6	40.2	49.9	64.5	69.8	72.8	71.7	69.4	58.8	48.6	37.7		
		MCDB	36.7	36.1	43.8	56.5	73.0	78.0	80.4	78.4	75.8	63.2	50.6	38.7		
	5%	WB	32.2	32.6	37.3	46.6	60.7	67.6	70.8	69.9	66.6	55.6	44.6	34.9		
		MCDB	33.5	34.2	40.6	54.3	67.9	74.8	77.5	76.2	72.2	59.6	47.0	36.0		
	10%	WB	28.2	30.0	34.8	43.7	57.2	65.4	68.9	68.2	63.8	52.2	41.7	32.7		
		MCDB	29.4	31.7	37.7	50.0	66.0	72.3	75.2	74.5	69.3	55.4	44.0	33.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.9	15.3	14.7	14.8	17.7	17.5	16.9	16.9	16.9	12.8	10.6	11.3		
		MCDBR	14.8	15.0	17.5	24.4	26.5	23.0	20.9	21.0	21.4	19.1	15.9	11.6		
		MCWBR	14.5	13.0	12.5	14.8	13.4	11.4	9.8	10.0	11.8	13.1	13.0	11.3		
		MCDBR	14.8	13.5	16.0	20.7	22.1	19.5	17.6	16.9	18.6	17.4	14.8	11.2		
	5% WB	MCWBR	14.9	12.4	12.7	14.3	13.7	11.8	9.8	9.8	12.0	13.1	13.1	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.283	0.325	0.350	0.365	0.402	0.412	0.394	0.368	0.337	0.307	0.292		
	taud		2.316	2.249	2.221	2.348	2.398	2.291	2.283	2.336	2.401	2.481	2.471	2.278		
	Ebn at Noon		265	281	284	287	285	274	269	270	267	260	248	234		
	Edh at Noon		23	31	37	37	36	41	40	37	31	25	21	22		
(44) All-Sky Solar Radiation	RadAvg		446	733	1117	1409	1606	1751	1716	1490	1170	684	396	300		
	RadStd		62	90	91	121	167	110	168	94	99	57	47	30		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page